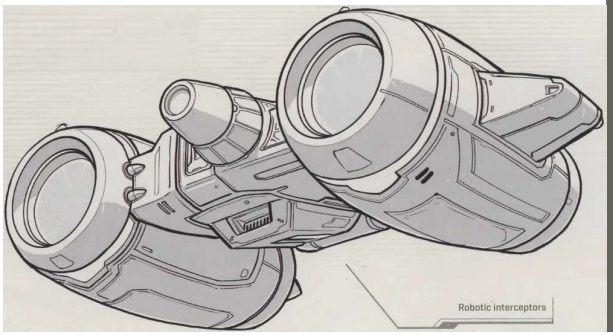
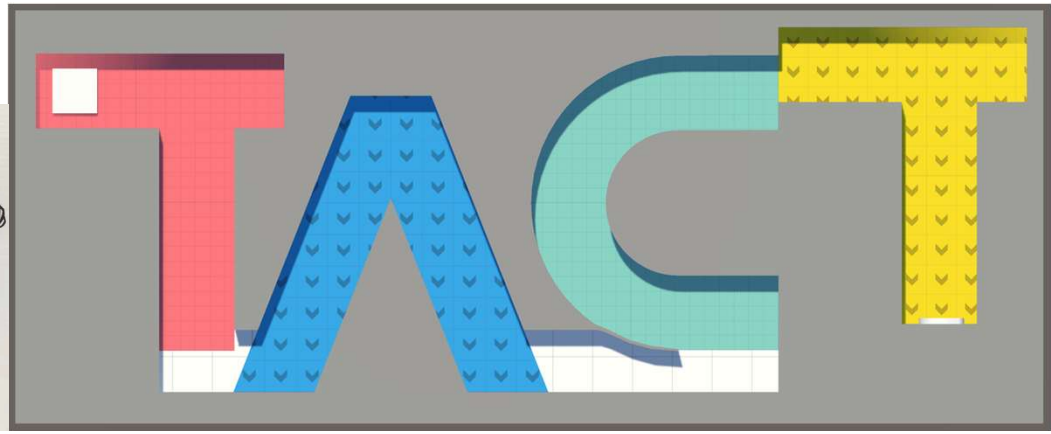
TA session 05

Project 2

Project2

- **Automatic control**
  - Automatically control the drone to reach the destination
  - Same track, Same rule
    - But no controller
    - Collision will add +5s penalty time
    - fastest team will get +5 bonus for final score
  - Same actuators (engine, airbrake, servo motor)
    - Also same frames + cylinder frame





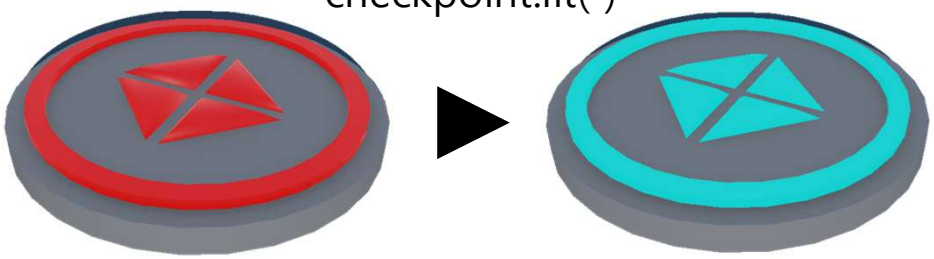
Project2

- **New parts**
  - **Drone body** (10kg > **1kg**)
  - **GPS** (track current position and position of checkpoints)

    - **Checkpoints** (place on map, use as temporal destination or just location beacon)
  - **IMU** (track 6dof Acceleration)

- **New rules**
  - Smaller Max thrust sum (60N > **30N**)
  - Airbrake has become weaker (coeff 10 > **1**)

checkpoint.lit( )



Project2

- **New control system**: DroneControlUnit.cs
  - Can use list of GPS, IMU
  - **Position**: GPS[index].currentPos (Vector3)
  - **Checkpoint position**: GPS[index].checkpoints[index].transform.position (Vector3)
    - Each GPS can have List of checkpoints
  - **Acceleration**: IMU[index].accel (Vector3)
  - **Angular acceleration**: IMU[index].ang_accel (Vector3)
  - **Time**: Time.FixedDeltaTime (float)

Project2

- **New Data collection Skill**: save data to .csv
  - Save your data for further control system analysis
    - Position vs time (refer to TrajectoryRecorder.cs)
    - Position of checkpoints (refer to PositionLogger.cs)
    - Velocity vs distance for stop
    - Time required to turn the body of drone to certain angle
    - Load data (refer to csvLoader.cs)
- **Quick Debug**
  - Debug.Log( );
  - Debug.LogError( );
  - Debug.LogWarning( );

	A	B	C
1	velocity	distance	coefficient
2	10	1.488738	0.148874
3	20	1.286263	0.064313
4	30	1.838585	0.061286
5	40	2.34906	0.058727
6	50	2.826996	0.05654
7	60	3.296814	0.054947
8	70	3.75592	0.053656
9	80	4.200928	0.052512

Restrictions

If any of the restrictions are violated, the performance report will be disqualified. (0 points).

- **Overall**
  - Do not drive off the given track (through a wall or across the y-axis)
  - If you want to use external libraries or assets, be sure to get prior approval.
  - You must not modify the structure, size, or components of the track or time, physics system in Unity.
- **Spacecraft**
  - Total size must fit in 9 * 9 * 5m box.
  - Only one **drone body** part can be used, and it must be the highest parent in the hierarchy.
  - Do not manipulate the size, weight or performance of the parts. (except frames)
  - The total thrust of all engines used in a rocket must be less than **30**.
  - A completed drone must not have any disconnected parts.
  - Except for the frame, no parts can overlap. **(servoarm will count as a frame)**
    - Frame-frame (o)
    - Frame-part (o)
    - Part-part (x)
  - Design so that interference does not occur even when the part is rotated by a servo motor.
- **Control**
  - Drone must run without any control except for start.

Files to be submitted

1. **Performance report** (same template with project 1)
   - Performance record
   - Participation details
   - Discussion
2. **Performance video**
   - Proof of your performance report.
   - full view of Unity scene and game window recorded from starting line to finish line.
     - Screen recording recommended
3. **Resource file Unity package**
   - Must contain full assets of the unity project

- **The driving records in the performance report must be reproducible using only submitted unitypackage file. (If not, please ask to TA first.)**

Evaluation criteria

- **Performance Report** (10 points, Team score)
  - Final Time = Time + Number of Collisions * 5 seconds
  - If a collision occurs, no further collisions will be detected for 5 seconds.
    - Detailed scoring criteria TBD
  - **The highest-scoring team: 5 additional points** will be awarded to the final grade.
    - This team's spaceship design, control system code, and video will be made public.

Evaluation criteria

- **Presentation** (15 minutes, 15 points, Individual score)
  - Presentation schedule is announced on the blackboard.
    - 11/18 (Tue), 11/20 (Thu)
    - Presentation order TBD
  - Each team will present their performance report in the professor's office.
  - Each team member present their role (All team members must present)
    - Performance record
    - Drone design
    - Control system design
    - **Iterations and Data analysis**